

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ CAUTION ⚠

Use caution when draining oil or replacing filters that the oil is not spilled or drained into the bilge area. The oil and oil filters must be disposed of in accordance with local environmental regulations.

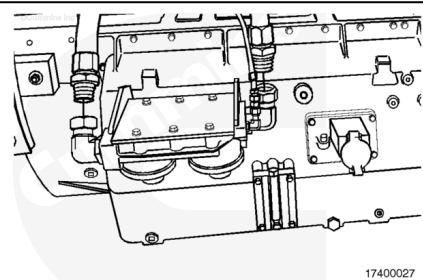
- Remove lubricating oil filters. Refer to Procedure 007-013 in Section 7.
(</qs3/pubsys2/xml/en/procedures/20/20-007-013-tr.html>)

Remove

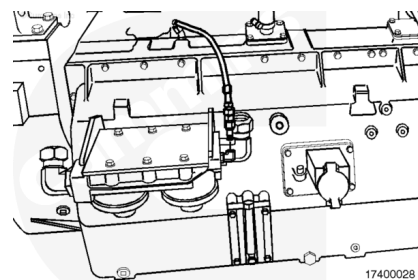
Rail Applications

Note : A remote oil filter is used as an example in this procedure. The filter can be mounted in a different location depending on the original equipment manufacturer (OEM).

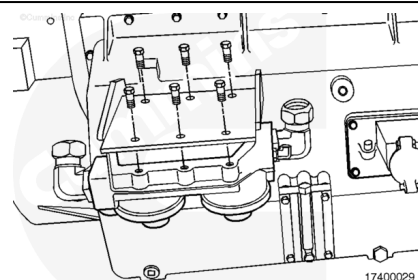
Disconnect the inlet and outlet hoses from the lubricating oil filter head.



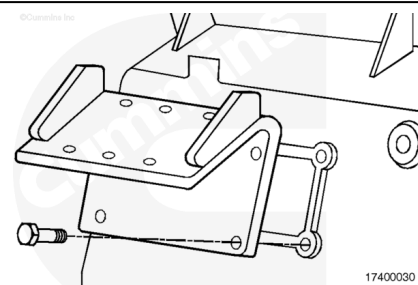
Disconnect the bypass oil hose from the lubricating oil filter head.



Remove the capscrews and the lubricating oil filter head from the bracket.



Remove the capscrews and the lubricating oil filter head bracket.



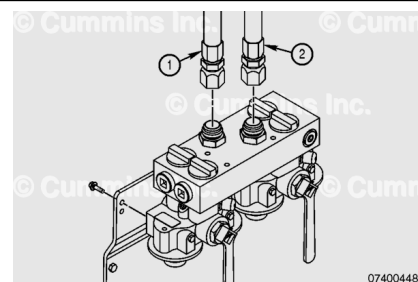
Marine Applications

Note : The remote mounted lubricating oil filter head can be mounted in different locations depending on the OEM.

Disconnect the inlet (1) and outlet (2) hoses from the lubricating oil filter head.

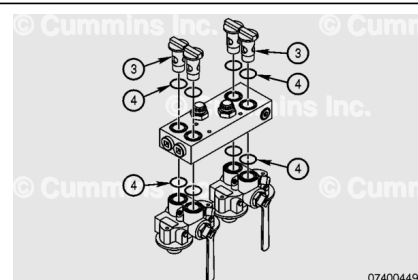
Remove the capscrews and the lubricating oil filter head from the bracket.

Note : See equipment manufacturer service information for bracket related disassembly details.



Remove the oil manifold connectors (3). Remove the oil manifold from the lubricating oil filter head.

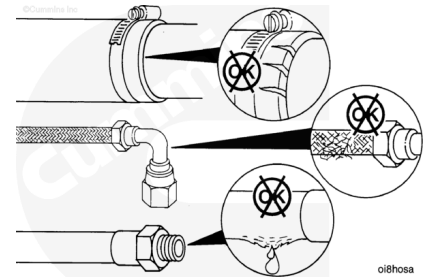
Discard the 8 o-rings (4).



Inspect for Reuse

Check lines and hoses for cracks.

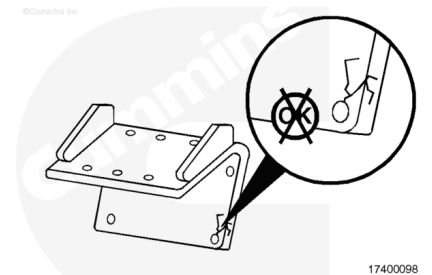
If the lines or hoses are cracked, they **must** be replaced.



Note : Remote mount lubricating oil filter head brackets may look different than in the illustration shown. The brackets are OEM dependent.

Inspect the lubricating oil filter head bracket for cracks and other damage.

If the lubricating oil filter head bracket is cracked or damaged, it **must** be repaired or replaced.

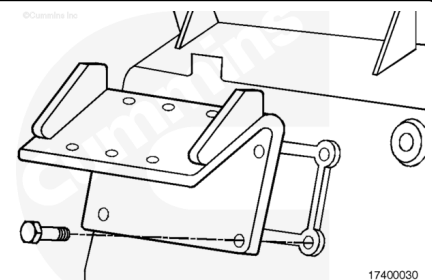


Install

Rail Applications

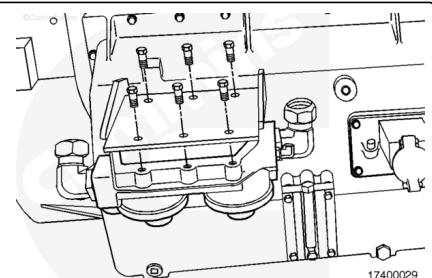
Install the lubricating oil filter head bracket and capscrews. Tighten the capscrews.

Use the following procedure for the torque value. Refer to Procedure 018-009 in Section V.
(/qs3/pubsys2/xml/en/procedures/99/99-018-009.html)



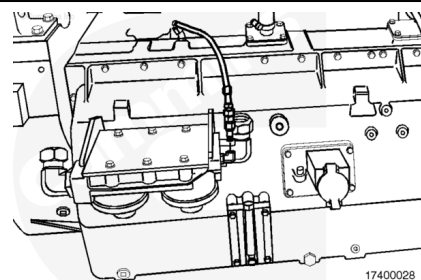
Install lubricating oil filter head and capscrews. Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]



Connect the bypass oil hose to the lubricating oil filter head.

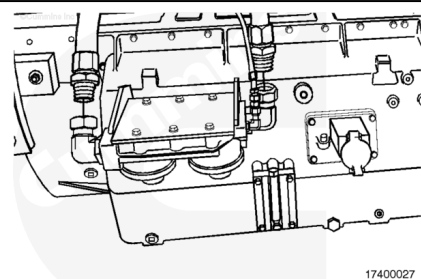
Torque Value: 26 n•m [19 ft-lb]



Connect the inlet and outlet hoses to the lubricating oil filter head.

Torque Value: 129 n•m [95 ft-lb]

Note : Make sure port labeled out on block connects to inlet port on filter head, and outlet port on filter head connects to inlet port on block.



Marine Applications

Install the lubricating oil filter head bracket and capscrews. Tighten the capscrews.

See equipment manufacturer service information for bracket location and installation details. Use the following procedure for the torque value.

Refer to Procedure 018-009 in Section V.

(/qs3/pubsys2/xml/en/procedures/99/99-018-009.html)

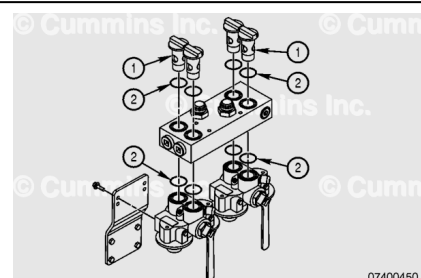
Install the lubricating oil filter head to the oil manifold, using the oil manifold connectors (1) and new o-rings (2).

Torque Value: 135 n•m [100 ft-lb]

Install the lubricating oil filter head to the bracket. Tighten the capscrews. See equipment manufacturer service information for bracket location and installation details. Use the following procedure for the torque value.

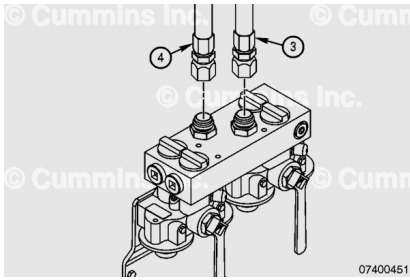
Refer to Procedure 018-009 in Section V.

(/qs3/pubsys2/xml/en/procedures/99/99-018-009.html)



Connect the inlet (3) and outlet (4) hoses to the lubricating oil filter head.

Note : Make sure port labeled out on block connects to inlet port on filter head, and outlet port on filter head connects to inlet port on block.



Finishing Steps

- Install the lubricating oil filters. Refer to Procedure 007-013 in Section 7.
(</qs3/pubsys2/xml/en/procedures/20/20-007-013-tr.html>)
- Operate the engine. Check for leaks.